

When the Story Doesn’t Match the Math: Runtime Verification of LLM-Narrated SHAP Explanations

Abstract

When a production ML system must explain a prediction to a person — a declined loan, a flagged network connection — it increasingly hands the job to a large language model (LLM): the underlying model’s feature attributions (e.g., its SHAP values) go in as context, and a fluent plain-English rationale for the applicant, operator, or auditor comes out. That last step is unchecked: the sentence *sounds* like the math, but nothing forces it to *match* the math. We document a silent failure mode we call **valence–magnitude inversion**: the narrative orders features by tone (good news first) instead of importance, so every stated fact is correct while the ranking is wrong — invisible to checks that only test factuality. Because the ground-truth SHAP vector is known at generation time, faithfulness can be verified automatically and nearly for free; we propose a per-output runtime contract — rank agreement between the narrative’s mention order and the true $|\text{SHAP}|$ order — with a no-human fallback ladder and a tunable flag rate. We evaluate seven current models drawn deliberately from the fast, cost-efficient “workhorse” tier that hot-path narration would actually use, on (a) synthetic loan-decision SHAP vectors under four prompt conditions and (b) *real* per-alert TreeSHAP values from a network-intrusion detector (50 adversarially selected “trap” alerts per model, $n=50$ scoreable for every model). At baseline, 52–76% of loan narratives mis-order the features and mean rank correlation with the true order is ≈ 0 ; the good-news-first share of trap narratives is largest for the strongest models (up to 50%). An explicit ordering instruction splits the fleet three ways: *robust* compliers (Kimi K3, Qwen 3.7 Max, GPT-5.6 Luna, Gemini 3.6 Flash — $\rho=1.00$ that survives a competing “be warm” instruction), *partial* compliers (GLM-5.2, Claude Haiku 4.5 — improve, then degrade again under warmth), and *non-compliers* (SmolLM3-3B — unchanged). On real network SHAP the baseline inversion rate spans 0% to 86% across the same seven models, and individual models flip between domains — SmolLM3-3B is perfectly faithful on network alerts yet 76% inverted on loans; Claude Haiku 4.5 shows the same flip. Faithfulness is unpredictable from model identity, tier, vendor, or even the same model’s behavior in another domain; narrated explanations need per-output verification, not just better prompting.

1 A silent failure mode in narrated explanations

Ask an LLM to turn a SHAP vector into a customer-facing sentence and it will oblige, fluently. In our own demo: “*The primary driver for your approval is your excellent Credit Score. Your Annual Income also played a positive role, though your high Debt Ratio slightly decreased your score.*” Every fact is correct. But the attributions were $(+0.45, +0.25, -0.35)$: by magnitude, Debt Ratio outranks Annual Income. The model ordered features by *valence* — good news first — not by importance. A reader who trusts the order of the prose walks away with the wrong causal picture, and no factuality-based detector fires, because nothing was fabricated.

Using an LLM as the explanation interface for ML predictions — the layer that turns a model’s decision into something a person can read — is spreading across credit scoring, security operations, and industrial monitoring. Under the hood, the LLM is typically prompted with the underlying model’s feature attributions, and its narrative is precisely what a regulator, operator, or applicant reads. Unlike open-ended generation, the setting carries its own ground truth: the attribution vector is available at generation time, so verification is nearly free. It simply is not being done.

Contributions. (1) We name and demonstrate *valence–magnitude inversion*, a faithfulness failure factuality checks cannot catch; (2) a runtime contract check (rank agreement with the true $|\text{SHAP}|$ order) with a no-human fallback ladder and tunable alert rate; (3) preliminary evidence across seven workhorse-tier models and two domains — one with real per-alert TreeSHAP — that inversion is systematic, that instruction-compliance splits robust/partial/none by model, and that faithfulness swings across domains *within* a model, so no static property predicts whether a given output is faithful.

2 Valence–magnitude inversion

Let $\phi \in \mathbb{R}^d$ be the SHAP values for one prediction and let π^* rank features by $|\phi_j|$ descending. A narrative induces its own ranking $\hat{\pi}$, operationalized as first-mention order. *Inversion* occurs when $\hat{\pi} \neq \pi^*$; *valence–magnitude inversion* is the specific, directional case where $\hat{\pi}$ sorts positive contributions ahead of larger-magnitude negative ones (“good news first”) — systematic rather than random, driven by the politeness conventions of instruction-tuned models, and consequential wherever stated priority matters (adverse-action notices, incident triage, model-risk reviews).

3 A runtime faithfulness contract

Rank check (confident wrongness). Extract $\hat{\pi}$ from the narrative (first-mention order over a known feature vocabulary; nothing beyond pattern matching) and compute Spearman’s $\rho(\hat{\pi}, \pi^*)$ over the mentioned features; unmentioned features count against a coverage requirement.

Fallback ladder (no human rater). Flagged narratives descend: regenerate once $\rightarrow$ templated explanation (always faithful, merely uglier) $\rightarrow$ raw attribution table. The gate changes what the existing operator sees, not who exists. (Where token logprobs are available, an uncertainty tripwire on generation entropy can be layered on top; we do not evaluate it here.)

4 Preliminary experiments

Model selection — why the “workhorse” tier. Narrating an attribution is a per-decision, hot-path task: in deployment it runs once for every loan adjudicated or every alert raised — potentially millions of calls a day — under tight latency and unit-cost budgets. Those economics rule out flagship reasoning models for the narration step and select exactly the tier we study: each vendor’s light, fast, cost-efficient offering — Gemini 3.6 Flash, Claude Haiku 4.5, GPT-5.6 Luna, GLM-5.2, Kimi K3, Qwen 3.7 Max — plus a small open-weight model (SmolLM3-3B) standing in for data-sensitive finance and SOC settings that cannot ship records to a hosted API. The failure rates reported below are therefore the ones a real deployment of this pattern would encounter.

Setup. For each model we generate narratives for synthetic loan-decision SHAP vectors constructed so that tone order and magnitude order disagree (“trap” instances; 82 of 85), under four conditions: C0 no ordering instruction; C1 explicit “discuss factors by absolute contribution, most important first”; C2 = C1 + a competing “be warm and reassuring” style instruction; C3 = C1 with six features instead of three (85 generations per model, $n=25/25/25/10$ scoreable across C0–C3). SmolLM3-3B runs locally (T4); Gemini 3.6 Flash on its native API; the rest via OpenRouter. Gemini is a reasoning model: with default output budgets its internal thinking silently truncates the visible answer, so narration evals must budget tokens for thinking *plus* answer (we use 3072) — a methods trap worth flagging. We then wire the same scorer into an anomaly-detection pipeline (Random Forest + SMOTE on synthetic SDN traffic, holdout AUC ≈ 1.0) narrating *real* per-alert TreeSHAP values: top-5 drivers per alert, restricted to the trap alerts where tone order and magnitude order disagree ($\sim 39\%$ of detected anomalies arise as traps naturally), 50 alerts per model at C0, with regeneration on unscoreable outputs so every model reaches $n=50$.

Findings. (i) *Unfaithful ordering is the baseline norm for every model:* at C0, 52–76% of loan narratives mis-order features and mean ρ sits between -0.10 and 0.40 — prose order is essentially uninformative about importance. (ii) *The strongest models fail most politely:* the tone-ordered (good-news-first) share of trap narratives is highest for Kimi K3 (**50%**) and Gemini 3.6 Flash (42%) on loans, and for Gemini (**54%**) and Kimi (50%) on network alerts. A typical Kimi miss: for (Recent Inquiries -0.47 , Credit Score $+0.40$, Loan Amount $+0.24$) it leads with “your strong credit score” and buries the larger negative inquiries last. (iii) *Instruction-compliance splits the fleet three ways, and that is the point: robust* — Gemini 3.6 Flash, Kimi K3, Qwen 3.7 Max, and GPT-5.6 Luna become faithful under the ordering instruction (C1 $\rho=1.00$, 0% inverted) and *stay* faithful under the competing warmth instruction and six features; *partial* — GLM-5.2 ($\rho=0.65$, 22% still inverted at C1) and Claude Haiku 4.5 ($\rho=0.82$, 20%) improve but degrade again under warmth (33% and 40%); *none* — SmolLM3-3B is unchanged (76% inverted at both C0 and C1). The split cuts across vendors and sizes and is invisible from outside; only a runtime check reveals whether *this* output is faithful. (iv) *On real attributions, model identity dominates — and flips across domains* (Table 2): baseline

Table 1: Loan-decision SHAP (synthetic vectors): rank faithfulness over scoreable narratives, seven workhorse-tier models $\times$ four conditions. “% inv.”: mention order $\neq$ |SHAP| order (strict); “% tone”: trap narratives following exactly good-news-first order. Note on C3: with six features an *exact* magnitude match is rare, so strict % inv. can be high even when ρ is high (Haiku 100%/ $\rho=0.79$; Luna 10%/ $\rho=0.99$) — we report both for this reason.

Model	Condition	n	mean ρ	% inv.	% tone (traps)
SmolLM3-3B	C0	25	−0.10	76%	17%
SmolLM3-3B	C1_order	25	−0.10	76%	17%
SmolLM3-3B	C2_warm	25	−0.12	80%	21%
SmolLM3-3B	C3_6feat	10	0.28	100%	20%
Gemini 3.6 Flash	C0	20	0.18	70%	42%
Gemini 3.6 Flash	C1_order	25	1.00	0%	0%
Gemini 3.6 Flash	C2_warm	25	1.00	0%	0%
Gemini 3.6 Flash	C3_6feat	10	1.00	0%	0%
GLM-5.2	C0	22	0.02	73%	24%
GLM-5.2	C1_order	23	0.65	22%	5%
GLM-5.2	C2_warm	24	0.50	33%	9%
GLM-5.2	C3_6feat	10	0.70	60%	0%
Claude Haiku 4.5	C0	25	0.40	52%	25%
Claude Haiku 4.5	C1_order	25	0.82	20%	8%
Claude Haiku 4.5	C2_warm	25	0.64	40%	21%
Claude Haiku 4.5	C3_6feat	10	0.79	100%	10%
Kimi K3	C0	23	0.26	74%	50%
Kimi K3	C1_order	25	1.00	0%	0%
Kimi K3	C2_warm	22	0.93	5%	5%
Kimi K3	C3_6feat	9	1.00	0%	0%
Qwen 3.7 Max	C0	25	0.28	72%	29%
Qwen 3.7 Max	C1_order	25	1.00	0%	0%
Qwen 3.7 Max	C2_warm	25	1.00	0%	0%
Qwen 3.7 Max	C3_6feat	10	1.00	0%	0%
GPT-5.6 Luna	C0	25	0.26	76%	29%
GPT-5.6 Luna	C1_order	25	1.00	0%	0%
GPT-5.6 Luna	C2_warm	25	1.00	0%	0%
GPT-5.6 Luna	C3_6feat	10	0.99	10%	0%

Table 2: Network-security case study: *real* per-alert TreeSHAP from the RF anomaly detector, top-5 drivers, 50 trap alerts per model, C0 (no ordering instruction), $n=50$ scoreable for every model. Baseline inversion spans 0–86% across the same seven models; the two most faithful narrators here (SmolLM3-3B, Claude Haiku 4.5) are the two smallest models — and both invert heavily on the loan vectors of Table 1.

Model	n	mean ρ	% inv.	% tone	coverage
SmolLM3-3B	50	1.00	0%	0%	0.98
Gemini 3.6 Flash	50	0.85	80%	54%	1.00
GLM-5.2	50	0.92	42%	18%	1.00
Claude Haiku 4.5	50	0.99	12%	12%	1.00
Kimi K3	50	0.90	60%	50%	1.00
Qwen 3.7 Max	50	0.81	86%	10%	1.00
GPT-5.6 Luna	50	0.92	52%	38%	1.00

inversion on real per-alert TreeSHAP spans 0% (SmolLM3-3B, $\rho=1.00$) to 86% (Qwen 3.7 Max) across the same seven models, with no size or vendor pattern — the two most faithful network narrators are the two *smallest* models. The same SmolLM3-3B that is perfectly faithful here is 76% inverted on the loan vectors; Claude Haiku 4.5 shows the same flip (12% vs. 52%). Faithfulness is a property of the (model, domain, instance) triple, not of the model.

5 Discussion

Prompting is mitigation, and an unreliable one: the ordering instruction did nothing for SmolLM3-3B and only partly worked for GLM-5.2 and Claude Haiku 4.5, with warmth eroding both. Nor is faithfulness fixed *within* a model: SmolLM3-3B is 76% inverted on the loan vectors but 0% on the network detector’s real TreeSHAP values, and Haiku shows the same domain flip — unpredictable from model identity even at fixed prompt. Because the models under study are exactly the fast workhorse tier that per-decision narration deploys, these are the operative failure rates, not a worst case. A dependable system therefore needs to know whether *this* output is faithful — and the check costs one regex pass and one statistic per output, doubling as the regression test for any prompt or model change. **Limitations.** First-mention order proxies claimed importance; the loan vectors are synthetic (the network case study, by contrast, scores real per-alert TreeSHAP from a detector

trained on synthetic SDN traffic); the uncertainty tripwire is proposed but not evaluated here; and stress conditions (C1/C2) have so far been run on the loan domain only, with the network-domain replication in progress.

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